

Google develops a framework to bring C++ closer to Python

Google has built an intrinsic open source project to let developers create C++ wrapper generators for Python. Called CLIF, the new framework automatically generates C++ library bindings for multiple languages, though it initially supports Python 2 and 3.

CLIF consists of four different parts, namely parser, matcher, generator and runtime. It leverages the LLVM compiler framework to convert a language-friendly C++ API description into the language-agnostic internal format. For collecting any typed information, the framework parses selected C++ headers with Clang.

Apart from parsing and matching the typed data, CLIF is capable of emitting C++ source code for a wrapper by using its generator component. The same wrapper is then used in the runtime to perform type conversions.

Google has provided elements such as its protobuf (for inter-process communication between



the CLIF front-end and back-end) and the CMake build system. Going forward, the framework is likely to become smarter with features such as inheritance. A feature to map exceptions in two languages is also expected to be added in a later version.

Meanwhile, you can access the CLIF code from a GitHub repository. The online listing explicitly mentions that the project has no formal link with Google.

“The folks at Google, along with us at the MagPi, are really excited to see what projects you can create (or enhance) with this kit -- whether you are creating a voice-controlled robot or a voice interface that answers all your questions,” said Rob Zwetsloot, features editor for the MagPi magazine.

You need to use your Raspberry Pi 3 to begin the action with the AIY kit. Also, Google has provided access to the recently developed Google Assistant SDK and Google Cloud Speech API to let you easily build the next Amazon Echo or Google Home.

Microsoft's Imagine Cup (2017) India round winners declared

Microsoft recently announced the India winners of its Imagine Cup 2017 at a conference in New Delhi. The winners will represent the country at the worldwide finals in July at Seattle to compete against global talent for a prize worth US\$ 100,000.

The team, Content Holmes from BITS Pilani

has emerged as the winning team among the 27 finalists from the Indian region. It developed a one-click artificial intelligence (AI) based online parenting solution to help parents keep children secure from cyber bullying.

Started back in 2003, the 15th edition of Imagine Cup attracted participation from more than 1,600 colleges across the country. “We had got entries from 500-600 institutes last year, but over 1,600 colleges have participated this time,” said Narendra Bhandari, general manager of the developer experience and evangelism, Microsoft India.

Bhandari, who leads the growth of developer engagements by Microsoft in the country, told *Open Source For You* that of the 27 participating teams, 11 were led by women. “This is definitely heartening, if not surprising,” he said.

Microsoft had four judging panels to pick the winners. Each of the panels comprised different jury members, including engineers, venture capitalists/accelerators, startup founders and academicians. And there was only one Microsoft employee in the panels.

“We have expanded the juries. They bring a different perspective when judging the ideas, and each of the widened jury panels is giving participants’ feedback to improve,” said Bhandari.

The aim of the Imagine Cup is not just to create startups but also to encourage thinking skills. Instead of focusing primarily on a proprietary solution, Microsoft is open to considering developments irrespective of the platform. “The message we give to people through the Imagine Cup is—any developer, any app and any platform,” Bhandari stated.

When asked about what message developers need to receive from a programme like Imagine Cup, Bhandari told *Open Source For You* that this is the best time to innovate. “You have all kinds of mentorship available,” he added.

Apart from Content Holmes, Mi-amigos of Chameli Group of Institutes, PocketConfident AI of IIT, Hyderabad, and White Cane of IIT, Kharagpur have emerged as the three runner-ups. Three of the four outstanding projects are based on AI technologies.

